

Document 31. Worked CSQ Solutions (4 annotated cases)

Four full Case Study walkthroughs. Each illustrates the protocol from Doc 26 with the mark-scheme-aligned language from Doc 25.

Section A. CSQ 1 — Carbon Tax and Singapore's Energy Transition

Stylised extract (paraphrased). Singapore raised its carbon tax from S\$5 to S\$25 per tCO₂e in 2024. Energy prices for businesses rose; some manufacturers relocated production to Malaysia. The Green Plan 2030 commits Singapore to net-zero by 2050.

Q1(a) Describe the trend in Singapore's carbon tax from 2019 to 2030. [2]

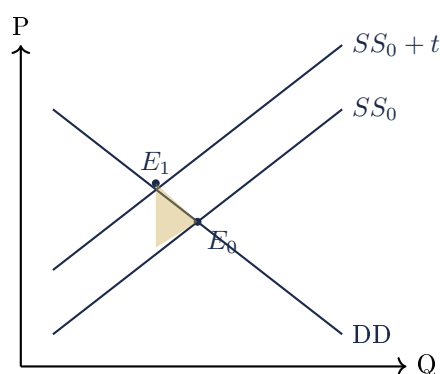
The carbon tax was constant at S\$5 from 2019 to 2023, then rose sharply to S\$25 in 2024, with announced increases to S\$45 by 2026–27 and S\$50–80 by 2030. *[Trend + magnitude + turning point]*

Q1(b) Explain how a rise in the carbon tax affects domestic firms' costs of production. [4]

A carbon tax raises the input cost of carbon-intensive energy (electricity, fuel). Firms' total variable costs rise proportionally with their emissions intensity. Consequently, marginal cost (MC) and average cost (AC) curves shift upward. Firms either pass costs on through higher prices (if PED is inelastic) or absorb them (lowering profit margins). *[Mechanism with cost-curve language]*

Q1(c) Using a diagram, analyse how the carbon tax affects the market for electricity in Singapore. [8]

The carbon tax raises MC for fossil-fuel-fired power generation. Supply curve shifts up by the tax amount ($SS_0 \rightarrow SS_0 + t$). *[State the shift]*



At the new equilibrium E_1 , consumer price rises from P_0 to P_1 and quantity falls from Q_0 to Q_1 . Consumer surplus falls. Producer surplus also falls (price received net of tax is below P_0). Government revenue equals $t \times Q_1$. Two DWL triangles emerge — representing the welfare lost on units no longer produced. *[Welfare analysis identifies winners / losers]*

In Singapore's context, the inelastic short-run demand for electricity means the immediate quantity reduction is modest; longer-run effects come via firm relocation, energy efficiency investments, and substitution toward solar / hydrogen. *[Context anchor with elasticity reasoning]*

Q1(d) Assess whether the carbon tax is the most appropriate policy to achieve Singapore's net-zero target. [10]

The carbon tax is well-suited because: (i) it internalises the externality via a Pigouvian mechanism; (ii) the escalator (S\$5 $\rightarrow$ S\$80) gives firms a predictable signal for capital planning; (iii) revenue funds R&D into clean technology; (iv) market-based design preserves productive efficiency by allowing firms to choose the cheapest abatement route. *[Multi-point case for the policy]*

Limitations: (i) current rate is below the IPCC's social cost of carbon estimate (US\$50–150/tCO₂e); (ii) coverage excludes small emitters and households, leaving large emissions un-priced; (iii) carbon leakage

risk — the extract notes manufacturers relocating to Malaysia; (iv) regressive distributional impact on low-income households via utility bills. *[Each limitation backed up]*

Alternatives: (i) tradable permits cap emissions directly but require active price discovery; (ii) regulation (mandatory emissions ceilings) is direct but inflexible; (iii) direct provision (state solar farms) speeds the transition but at fiscal cost; (iv) subsidies for clean tech complement the tax. *[Three alternatives evaluated]*

Stance: the carbon tax is appropriate as the *anchor* of Singapore's net-zero strategy, but cannot deliver net-zero alone. The Green Plan 2030 explicitly pairs it with solar deployment targets, EV adoption schemes, building efficiency standards, and the Climate Friendly Households Programme. *[Stance + synthesis with policy mix]*

Conclusion: the policy is well-designed but insufficient in isolation. The 2026–27 escalator must reach closer to international SCC estimates, and complementary policies must accelerate, for Singapore to credibly meet its 2050 target. *[Forward-looking conclusion]*

Section B. CSQ 2 — Inflation and MAS Response (2022–23)

Stylised extract. Singapore's core inflation peaked at 5.5% in January 2023, driven by imported energy and food prices. MAS tightened policy five times. Households received an Assurance Package totalling S\$8 billion.

Q2(a) Define core inflation. [2]

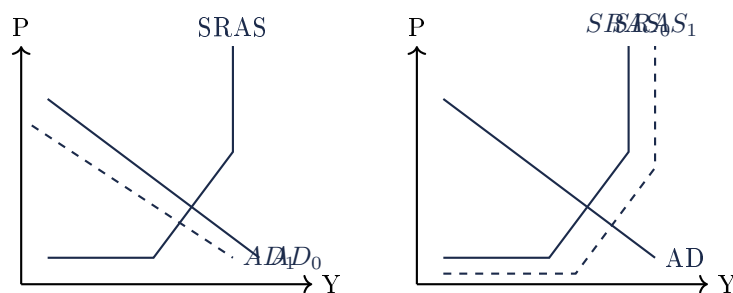
MAS Core Inflation is the year-on-year change in the Consumer Price Index excluding accommodation and private transport, used by MAS as its primary inflation indicator. *[Exact definition]*

Q2(b) Explain two reasons for the rise in Singapore's core inflation in 2022. [4]

First, imported cost-push: global energy and food prices rose sharply after the Russian invasion of Ukraine (Feb 2022). Since Singapore imports virtually all energy and most food, these costs feed directly into domestic prices via the SRAS curve. *[Mechanism + context]*

Second, a weakening US dollar / strong demand for imports increased the SGD cost of imports. Combined with re-opening demand (post-COVID), this contributed to both cost-push and demand-pull pressures. *[Second cause]*

Q2(c) Using AD-AS diagrams, analyse the effect of MAS's tightening on the Singapore economy. [8]



MAS tightening via S\$NEER appreciation has two transmission channels:

Channel 1 (AD). Stronger S\$ raises export prices in foreign currency and lowers import prices in S\$ terms. Net effect on $(X - M)$: ambiguous in theory but typically negative for trade-deficit sectors; positive for import-dependent consumption. Overall AD shifts left. *[Channel 1 with direction]*

Channel 2 (SRAS). Lower import costs reduce firms' input bill; SRAS shifts right. *[Channel 2]*
Combined effect: P falls (definitely); Y ambiguous (depends on relative shifts). Empirically, core inflation moderated from 5.5% to ~3% by end-2024 with little detectable growth cost, indicating SRAS channel dominated. *[Empirical resolution]*

Q2(d) Discuss whether MAS's tightening is the appropriate response to Singapore's 2022–23 inflation. [10]

Effectiveness: (i) MAS's framework is calibrated for imported cost-push — exchange rate appreciation directly addresses the import-price channel. (ii) The five-step tightening returned core inflation to target

by 2024. (iii) The framework avoids the trade-offs of interest-rate policy, which would have damaged the property and finance sectors.

Costs of tightening: (i) export competitiveness suffered for trade-exposed sectors; (ii) tourism and conference sectors faced higher SGD-denominated costs; (iii) FX volatility increased for Singapore-domiciled MNCs.

Complementary measures: the S\$8 billion Assurance Package directly cushioned low-income households via U-Save, CDC vouchers, and cash payouts. Foreign worker policy was tightened to dampen wage-cost spirals. *[Policy mix evidence]*

Alternative scenarios: (i) doing nothing would have allowed inflation expectations to de-anchor; (ii) using fiscal contraction would have hurt growth more; (iii) capital controls are off the table given Singapore's hub status.

Stance: MAS tightening was the appropriate response because the inflation was predominantly imported (60–70% of the 5.5% peak), and the framework's design directly targets this channel. *[Stance with magnitude]*

Conclusion: MAS's response was proportionate and effective. The 2024 normalisation (slight easing) reflects mission accomplished on the imported inflation channel. The remaining structural domestic inflation requires supply-side complements (productivity, food security). *[Conclusion + forward look]*